

Fergusson College (Autonomous), Pune
Dept. of Physics

Modeling A New $f(R)$ Theory of Gravity

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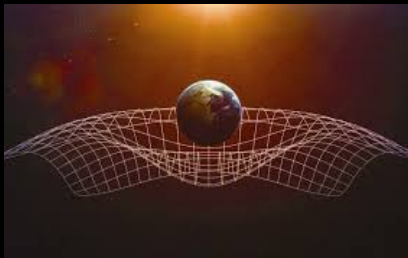
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Introduction

Gravity, on a cosmological scale, is governed by the Theory of General Relativity (GR):

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

In terms of matter-energy and curvature



Problem

$\Lambda \Rightarrow$ **Cosmological Constant**

It represents dark energy (in the Λ CDM model)

$\Rightarrow$ Responsible for the Cosmic Space-Time Expansion in Field Equations



Dark energy is still not fully understood, and there is a desire to find **alternative explanations** for cosmic acceleration, rather than relying on these unknown components.

Proposed Solution

Modified theories of gravity

Ones which can explain cosmic expansion **without invoking Λ**

- ▶ Dynamical Chern-Simons Modified Gravity
- ▶ $f(R)$ Theory of Gravity
- ▶ K-essence

Action & Least Action

Action: encapsulates the dynamics of a system

⇒ an integral over time

$$S = \int L dt$$

where L is Lagrangian of system, describes the energy of the system.

Least Action Principle: the physical path taken by a system is the one that makes the action minimum

$$\delta S = 0$$

Action in Field Theory

In field theory, the action is a functional of the field $\phi(x)$, given by:

$$S = \int d^4x \mathcal{L}(\phi, \partial_\mu \phi)$$

where $\mathcal{L}$ is the Lagrangian density, which describes the energy content and interactions of the system.

Action in Field Theory

The fundamental field in GR $\Rightarrow g_{\mu\nu} \Rightarrow$ describes the curvature of spacetime.

The **Lagrangian density for gravity** must satisfy the following conditions:

- ▶ **Generally Covariant:** The action should be independent of coordinate choice.
- ▶ **Contain No More Than Second-Order Derivatives:** To avoid ghost-like instabilities.
- ▶ **Reduce to Newtonian Gravity in the Weak-Field Limit** (weak grav. field and almost flat spacetime).

Action in Field Theory

The simplest scalar function that satisfies these conditions is the **Ricci scalar** R , which describes the curvature of spacetime. This leads to the **Einstein-Hilbert action**:

$$S_{EH} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} R$$

where:

- ▶ $g = \det(g_{\mu\nu})$ is the determinant of the metric tensor.
- ▶ R is the Ricci scalar, obtained from the **Ricci tensor** $R_{\mu\nu}$.
- ▶ $\kappa = \frac{8\pi G}{c^4}$ relates the action to Newton's gravitational constant G .

This action is the **most natural and minimal choice** for a covariant gravitational action.

$f(R)$ Theory of Gravity

We replace the Ricci curvature scalar R with a generalized function of the Ricci scalar $f(R)$ in the Einstein-Hilbert action.

$$S_{EH} = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} R$$

$\Downarrow$

$$S_{EH,f(R)} = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} f(R)$$

$f(R)$ Theory of Gravity

The total action (in Jordan Frame) is the net contribution from the Einstein-Hilbert action and the action for matter fields

$$S_{total} = S_{EH,f(R)} + S_{matter}$$

$$S_{total} = \frac{1}{2\kappa^2} \int d^4\mathbf{x} \sqrt{-g} f(R) + S_{matter}$$

$f(R)$ Theory of Gravity

We need to design a new $f(R)$

In unmodified action, $f(R) = R$

$\therefore$ In modified action $f(R) = R + [\text{correction terms}]$

Say, Correction Terms = $f_{corr}(R)$

Conditions for correction terms

- ▶ Should recover GR at high curvature, i.e. at $R \gg R_c \rightarrow f_{corr}(R) \approx 0$
- ▶ Modify low curvature to account for cosmic acceleration, i.e. at $R \ll R_c \rightarrow f_{corr}(R) \approx \text{constant}$ and the effective $f(R)$ should replace Λ to mimic the role of dark energy at low curvature

R_c is a characteristic curvature scale. It sets the scale at which modifications to General Relativity (GR) become significant.

Correction Terms Calculation

Because the model should reduce to GR at high curvature, we want additional terms to vanish at $R \gg R_c$; ($\frac{R_c^2}{R^2} \ll 1$)

For low curvature ($R \ll R_c$), the term should stabilize and give significant correction to GR; ($\frac{R_c^2}{R^2} \gg 1$)

The function that naturally satisfies this condition is the **inverse tangent** function

Correction Terms Calculation

Taking Taylor series expansion of $\tan^{-1}\left(\frac{R_c^2}{R^2}\right)$, Say $\frac{R_c^2}{R^2} = x$

For $x \ll 1$, $\tan^{-1}x \sim x$ and the higher terms vanish. Then using the approximation:

$$\cot^{-1}x = \frac{\pi}{2} - \tan^{-1}x$$

We get the **first part** of the correction term calculated to be:

$$\boxed{-\frac{\alpha}{\pi} R_c \cot^{-1}\left(\frac{R_c^2}{R^2}\right)}$$

This term modifies gravity at intermediate and low curvature scales. The function $\cot^{-1}$ smoothly interpolates between different curvature regimes

- ▶ At $R \gg R_c$, $f_{corr}(R) = \frac{\pi}{2} - \frac{R_c^2}{R^2}$
- ▶ At $R \ll R_c$, $f_{corr}(R) = -\frac{\alpha}{\pi} R_c \left(\pi - \frac{R_c^2}{R^2} \right)$

Correction Terms Calculation

Now we need a function behaves like the cosmological constant at very small R , leading to cosmic acceleration.

The form $1 - e^{-R/R_c}$ is chosen so that:

- ▶ For $R \gg R_c$, we have $e^{-R/R_c} \approx 0$, making this term negligible.
- ▶ For $R \ll R_c$, we have $e^{-R/R_c} \approx 1$, so the term approaches $-\beta R_c$, contributing to the effective dark energy.

This ensures a smooth transition between high and low curvature regimes.

Correction Terms Calculation

So the **second part** of the correction function is:

$$-\beta R_c \left(1 - \exp \left(-\frac{R}{R_c} \right) \right)$$

This exponential suppression term ensures a smooth transition at low curvature. It prevents singularities by keeping the function finite at very small R .

Behavior in Different Limits:

- **High curvature** ($R \gg R_c$):

$$\exp \left(-\frac{R}{R_c} \right) \approx 0$$

The term **vanishes**, making this correction irrelevant at large R .

- **Low curvature** ($R \ll R_c$):

$$\exp \left(-\frac{R}{R_c} \right) \approx 1 - \frac{R}{R_c} + \mathcal{O}(R^2)$$

The term **becomes important**, modifying gravitational dynamics.

The parameter β controls the strength of this term.

Correction Terms: Final Form

The final form of $f(R)$ then becomes:

$$f(R) = R - \frac{\alpha}{\pi} R_c \cot^{-1} \left(\frac{R_c^2}{R^2} \right) - \beta R_c \left[1 - \exp \left(-\frac{R}{R_c} \right) \right]$$

Using this into Total Action ($S_{EH} + S_{matter}$) and varying it under least action principle gives **modified field equation**:

$$f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} - \nabla_\mu \nabla_\nu f'(R) + g_{\mu\nu} \square f'(R) = \kappa^2 T_{\mu\nu}$$

A New Degree of Freedom

General Relativity has **2 degrees of freedom** — corresponding to the two polarization states of the gravitational wave.

Changing R to $f(R)$ brings modification to field equations, introducing a new kind of behavior that *wasn't present in GR*.

Therefore, we want to show that $f(R)$ gravity is mathematically **GR gravity plus a scalar field**.

That scalar field is represented as $\phi = f'(R)$, it obeys second order differential equations and changes dynamically.

So instead of only two degrees of freedom, we now have an *extra* degree of freedom - the scalaron degree of freedom.

Function of Scalaron in This Theory

In field theory, every field corresponds to a particle. So the scalar field corresponds to scalaron, the particle of the scalar field, with some scalaron mass.

This means the scalaron can mediate an additional “fifth force”, apart from the usual four fundamental forces, which should affect things like the motion of planets or objects in a lab.

But we don't see any such extra force in our solar system or on Earth.

So we utilize a physically motivated screening mechanism called **Chameleon Mechanism**.

Chameleon Mechanism

The chameleon mechanism proposes that:

The scalar field's mass **depends on the local environment** — specifically, on the **local matter density or curvature**.

Just like a chameleon changes its color to blend into its surroundings, this scalar field changes its mass to blend into the gravitational environment.

Chameleon Mechanism

In high-density regions (like Earth or the solar system):

- ▶ The scalar field becomes very massive.
- ▶ A massive scalar field has a very short range, so the fifth force it mediates dies out quickly.
- ▶ Effectively, the scalar field becomes undetectable, and GR is restored.

In low-density regions (like intergalactic space):

- ▶ The scalar field becomes very light.
- ▶ Its fifth force is long-range and can influence cosmic expansion or structure formation.
- ▶ It can drive accelerated expansion, mimicking dark energy.

Scalaron Mass

Taking trace (contracting its indices to obtain a scalar quantity) of the modified field equation gives:

$$f'(R)R + 3 \square f'(R) - 2f(R) = \kappa^2 T$$

and defining a scalar field

$$\phi = f'(R) = \frac{df(R)}{dR} = 1 - \frac{\alpha}{\pi} \frac{2R_c^3}{R^3 + \frac{R_c^4}{R}} + \exp\left(-\frac{R}{R_c}\right)$$

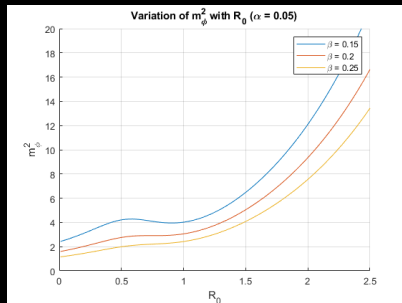
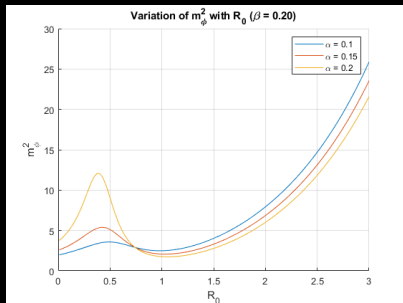
Using that into Klein-Gordon type equation, we get **the mass of scalar field under stationary condition** as:

$$m_\phi^2 = \frac{R_c \exp\left(\frac{R}{R_c}\right) \left(\pi (R^4 + R_c^4)^2 - 8\alpha R^5 R_c^3 \right) - \pi \beta (R + R_c) (R^4 + R_c^4)^2}{3\pi \beta (R^4 + R_c^4)^2 - 6\alpha R_c^4 \exp\left(\frac{R}{R_c}\right) (R_c^4 - 3R^4)}$$

for $R = R_0$ being background curvature

Scalaron Mass

m_ϕ depends on α, β and R_c . Setting $R_c = 1$, mass square of the scalar field as a function of the background curvature for different sets of values of α and β :



Conclusion

We have made **a new model of gravity by modifying Ricci scalar as a generalised function of Ricci scalar in EH Action.**

We have **calculated modified field equations** and calculated the **mass of scalar field** and saw **its variation with background curvature**

Future Scope

The following work can be done with this project:

- ▶ Solar System Tests of the Model
- ▶ Comparison of the model with other viable models
- ▶ Testing the model on gravitational waves by applying suitable constraints

Thank You!